Dimensionality reduction

Module 4 - Lesson 2

Overview

High-dimensional data is hard to visualize and noisy to model. Dimensionality reduction compresses features while keeping the essential structure.

Key Concepts

- PCA finds orthogonal directions of maximum variance and projects data onto them.
- The explained-variance ratio shows how much information each component carries.
- PCA is a linear method; t-SNE and UMAP capture non-linear structure for visualization.
- Reduction helps visualization, speed, and removing correlated noise.
- Keep enough components to retain most variance; the elbow of the scree plot guides you.
- PCA components are linear combinations of features; interpret with caution.

Example

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2)
projected = pca.fit_transform(X_scaled)
print(pca.explained_variance_ratio_)  # [0.52, 0.18] for two components

import matplotlib.pyplot as plt
plt.scatter(projected[:, 0], projected[:, 1], c=labels)
plt.show()
```

Summary

Dimensionality reduction compresses features for visualization and modeling. PCA captures linear variance; t-SNE/UMAP reveal non-linear structure. Always check explained variance.

Practice Checklist

- Run PCA and plot the explained variance for each component.
- Project a dataset to 2D and color by a cluster label.
- Explain the trade-off of keeping fewer components.